PeopleFlow: Floor-Plan-Aware Simulation for Building Evacuation Safety Assessment

Likith Sai Kunchi

***Abstract*—Evaluating evacuation performance in complex buildings requires analysis methods that capture the interaction between spatial configuration, pedestrian behavior, and operational disruptions. This paper introduces PeopleFlow, a geometry-informed computational framework designed to quantify evacuation effectiveness across diverse architectural layouts. The approach converts floor-plan representations into navigable simulation environments and applies continuous-space multi-agent modeling based on force-driven pedestrian dynamics. The system generates analytically meaningful indicators including total clearance duration, localized congestion intensity, persistence of bottleneck regions, and distribution balance across available exits.**

To enable consistent experimentation, the framework incorporates parameterized scenario configurations and experiment-traceable execution records. The evaluation includes a structured matrix of nine layout types across five operational conditions, resulting in 450 core simulation runs, complemented by 90 additional controlled experiments examining real-layout generalization and parameter sensitivity involving exit width, walking-speed preference, and occupant scaling. Observed results indicate strong dependence of evacuation performance on geometric topology and routing strategy, with clearance duration differences exceeding an eightfold range and substantial variation in congestion intensity across layouts.

The contribution of this work lies in establishing a repeatable engineering methodology for relative safety prioritization rather than deterministic prediction of individual trajectories. The proposed framework supports systematic what-if evaluation during early design stages and provides a computational basis for integration within digital twin environments for safety-oriented planning of high-occupancy infrastructure. The findings emphasize the importance of structured scenario evaluation to identify geometry-induced congestion risks and to support evidence-driven improvement of evacuation robustness.

***Index Terms*—Multi-agent simulation, evacuation modeling, crowd safety, building evacuation, digital twins, floor-plan analytics, safety engineering.**

I. Introduction

Building evacuation performance evaluation is a high-impact engineering task where static compliance checks often miss dynamic crowd effects under disruption. Historical emergencies consistently show that bottleneck formation, uneven exit utilization, and localized panic response can dominate outcomes even in code-compliant facilities. Conventional hand calculations remain useful for baseline checks, but they provide limited support for comparing design alternatives under multi-hazard and operational uncertainty [1], [2].

The need for early-stage design validation has increased with higher occupancy density in modern infrastructure. Although simulation enables repeatable comparative analysis before deployment, the central challenge is integrating architectural input, scenario control, execution, and interpretable outputs within one coherent computational workflow.

The present research introduces PeopleFlow as a geometry-informed evacuation performance framework with experiment-traceable execution. Rather than proposing a new pedestrian-physics law in isolation, the study develops a unified computational workflow for robust scenario-based evaluation across heterogeneous layouts.

The principal technical contributions can be summarized as follows:

- An experiment-traceable computational workflow linking floor plans, scenario definitions, simulation runs, and artifact-backed reporting.
- A geometry-informed modeling pipeline that converts architectural layouts into simulation-ready geometry.
- A density-sensitive path-selection comparison framework with explicit ablation evidence.
- A multi-layout statistical evaluation including a 450-run core matrix with cross-layout generalization analysis on airport, metro, and office plans.
- A statistically grounded sensitivity evaluation covering exit width, desired speed, and occupancy scaling.
- Analytically meaningful indicators for engineering evaluation, including clearance time, peak density, bottleneck persistence, and exit balance.

PeopleFlow is designed as a comparative decision-support workflow rather than an exact predictor of individual human trajectories.

II. Related Work

A. Crowd Modeling Approaches

Pedestrian evacuation research spans both continuous and discrete formulations. In force-based formulations, evacuees can be represented as self-driven particles whose motion evolves through interaction potentials capturing repulsion, attraction, and directional preference; this family remains a physically interpretable baseline for dense-crowd analysis [3], [4]. Cellular automata and floor-field variants offer efficient large-scale approximations but may introduce directional discretization artifacts near bottlenecks and tight corners [5], [6]. Comparative studies indicate that model suitability is task-dependent; for complex indoor geometries, continuous-space approaches often preserve turning behavior and local interaction structure more naturally [7]–[9].

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TABLE I
COMPARISON WITH EXISTING EVACUATION TOOLS

Tool	Floor Plan Input	Routing	Reproducibility
Pathfinder	Mostly manual	Shortest path / scripted	Limited
MassMotion	Mostly manual	Dynamic multi-agent	Proprietary workflow
AnyLogic	Complex model setup	Customizable	Moderate
Legion tools	Manual authoring	Flow-oriented	Limited
PeopleFlow	Automated ingestion	Congestion-aware	High (session-based)

B. Evacuation Simulation Systems

Commercial and academic evacuation platforms generally cover geometry authoring, route simulation, and report generation, but they differ substantially in automation depth, extensibility, and experiment traceability [10]–[12]. Behavior-focused studies further show that guidance logic, uncertainty handling, and panic-modulated decisions can materially alter evacuation outcomes [13], [14]. In many practical settings, heavy manual preprocessing and proprietary tooling still constrain transparent cross-layout benchmarking.

C. Digital Twin Safety Analysis

Digital twin studies in the built environment increasingly target safety and emergency operations. Prior work supports simulation-coupled twins for risk-aware planning, yet complete pipelines from floor-plan ingestion to repeatable quantitative evacuation indicators remain limited in many deployments [15]–[18]. Recent building-fire twin systems also demonstrate practical monitoring and control integration for emergency analytics, reinforcing the need for end-to-end experiment-traceable computational workflows [19], [20].

D. Research Gap and Positioning

The gap addressed in this paper is methodological: experiment-traceable, geometry-informed evacuation performance evaluation across diverse layouts and stress scenarios. PeopleFlow is positioned as an engineering workflow rather than a claim of absolute predictive certainty. Its novelty lies in coupling geometry extraction, session-managed simulation, scenario ablation, and interpretable reporting within one coherent computational workflow. A concise capability comparison with representative tools is provided in Table I.

III. METHODOLOGY

A. Workflow Pipeline

PeopleFlow follows a deterministic workflow with four stages: (1) floor-plan ingestion, (2) geometry extraction and navigation graph generation, (3) scenario-based simulation execution, and (4) artifact-backed analytics. Each run is stored with configuration metadata, seeds, and per-step metrics so results are replayable and auditable. The step-level simulation procedure is summarized in Algorithm 1.

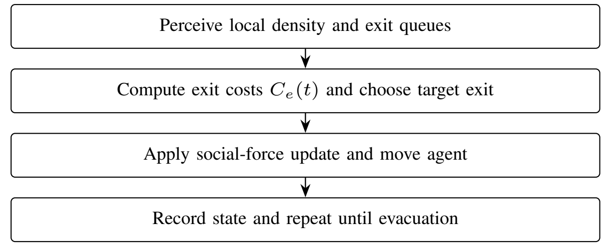

Fig. 1. Agent movement logic used in each simulation step.

B. Reproducibility Design

Each experiment is represented by a session artifact that captures:

- Input layout identifier and geometry hash.
- Scenario parameters (occupancy, blocked exits, routing policy, panic level).
- Random seed and simulation step size.
- Aggregated outputs and raw trajectory summaries.

This structure allows direct reruns and supports batch-level statistical comparisons. The modular organization also enables independent evaluation of routing logic without modifying geometric preprocessing components. Simulation configurations and synthetic benchmark layouts are packaged as reproducible experiment artifacts and can be released to support future comparative studies.

C. Agent Decision Logic

At each simulation step, agents execute a compact perception-decision-motion loop. Local density and queue conditions are sensed first, then exit costs are updated using Eq. (4). The lowest-cost exit is selected, social-force dynamics are integrated, and state variables are updated for the next step.

D. Computational Complexity

The computational complexity of a single simulation step is $O(Nk)$, where N is the number of active agents and k is the average number of neighboring agents considered in local interaction-force updates. Under practical density regimes, k remains locally bounded, giving near-linear scaling with population size for each step. For batch execution, total workload scales as $O(RNS)$, where R is the number of runs and S is the number of simulation steps per run. This complexity profile motivates the reproducible batch workflow used in our large-matrix experiments.

E. Scenario Matrix

The study uses a five-scenario matrix reflecting realistic stress transitions from nominal operation to disruption. Table II summarizes the definitions.

IV. PEOPLEFLOW ARCHITECTURE

A. System Components

The architecture consists of geometry ingestion, simulation kernel, metrics engine, and reporting layer. Geometry ingestion

TABLE II
SCENARIO DEFINITIONS

Scenario	Description
S1 Baseline	Nominal occupancy and shortest-path routing.
S2 HighOcc	Increased occupancy stress to test crowding sensitivity.
S3 Blocked	One major exit disabled to evaluate rerouting resilience.
S4 Routing	Congestion-aware routing policy activated.
S5 Panic	Stochastic panic variation in desired speed and interactions.

Algorithm 1 PeopleFlow Simulation Loop

```
Input: geometry G, scenario S, agents N, and step Δt
Initialize agents at valid non-overlapping spawn locations
Set routing and behavior parameters from S
while not all agents evacuated and t < T_max do
    for each agent i do
        Evaluate target exit based on the active routing policy
        Compute driving, interaction, and wall forces
        Update velocity and position via forward integration
    end for
    Update local density map and congestion indicators
    Record per-step metrics and event logs
    t ← t + Δt
end while
Aggregate run-level metrics and persist session artifacts
```

transforms floor plans into obstacle and exit primitives. The simulation kernel executes force-based motion and route updates. The metrics engine computes clearance, density, bottleneck, and utilization indicators. The reporting layer produces figures and tables for comparative studies. Figure 2 and Figure 3 summarize these system-level and workflow-level connections.

B. Session and Artifact Model

A session-centric design is used to support scientific traceability. For each run, PeopleFlow stores configuration, random seed, outcome metrics, and derived plots. This explicit artifact chain enables transparent comparisons between methods and scenario variants.

C. Analytics Outputs

Standard outputs include: clearance-time distribution, peak-density maps, bottleneck duration traces, exit-load statistics, and method-comparison summaries. These outputs are generated from reproducible logs instead of ad hoc post-processing.

V. MATHEMATICAL MODEL

A. Social Force Dynamics

Each agent i is modeled by a second-order force balance:

$$m_i \frac{d\mathbf{v}_i}{dt} = \mathbf{f}_i^{\mathrm{drv}} + \sum_{j \neq i} \mathbf{f}_{ij}^{\mathrm{int}} + \sum_{W} \mathbf{f}_{iW}^{\mathrm{wall}}. \quad (1)$$

The driving force is:

$$\mathbf{f}_i^{\mathrm{drv}} = m_i \frac{v_i^0 \mathbf{e}_i - \mathbf{v}_i}{\tau_i}, \quad (2)$$

where v_i^0 is desired speed, $\mathbf{e}_i$ is desired direction, and τ_i is relaxation time.

B. Density-Flow Relation

At a macroscopic level, local flow follows:

$$q = \rho v, \quad (3)$$

where q denotes pedestrian flow rate, ρ denotes local pedestrian concentration, and v represents average movement velocity. This relation is used for consistency checks between simulation outputs and empirical benchmarks [21], [22].

C. Exit Selection Cost Function

For congestion-aware routing, the exit cost at time t is:

$$C_e(t) = \alpha d_e(t) + \beta \rho_e(t) + \gamma h_e(t), \quad (4)$$

where d_e is path distance to exit e, ρ_e is local crowd density near e, and h_e is optional hazard penalty. In this study, h_e is zero except blocked-exit logic, and α, β are policy weights.

D. Statistical Metrics

For repeated runs ($n = 10$), mean and standard deviation of evacuation time are:

$$\bar{T} = \frac{1}{n} \sum_{k=1}^{n} T_k, \quad (5)$$

$$\sigma_T = \sqrt{\frac{1}{n} \sum_{k=1}^{n} (T_k - \bar{T})^2}. \quad (6)$$

Exit load imbalance is measured as:

$$I_{\mathrm{exit}} = \frac{1}{2N} \sum_{e=1}^{E} \left| n_e - \frac{N}{E} \right|, \quad (7)$$

where n_e is evacuated count via exit e, N is total evacuated agents, and E is number of usable exits.

VI. EXPERIMENTAL SETUP

A. Simulation Parameters

The core study executes 450 runs (9 layouts × 5 scenarios × 10 repetitions). To strengthen generalization and engineering interpretability, we additionally conduct 90 controlled runs: 30 supplementary real-layout runs (airport terminal, metro station, office building; baseline protocol), 30 exit-width comparison runs, and 30 desired-speed sensitivity runs. Key simulation parameters are shown in Table III. Nine layouts were selected to balance geometric diversity with computational feasibility while enabling statistically meaningful cross-layout comparison.

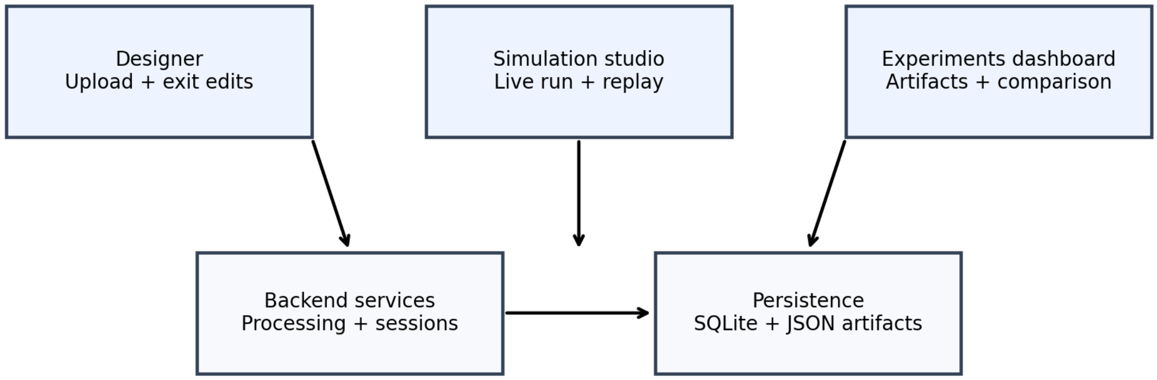

Fig. 2. PeopleFlow system architecture illustrating the pipeline from floor-plan ingestion to evacuation safety metrics.

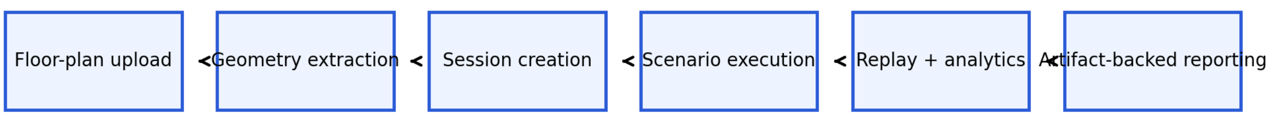

Fig. 3. Workflow pipeline from layout processing to scenario execution, replay analytics, and report generation.

TABLE III
SIMULATION PARAMETERS

Parameter	Value
Time step (Δt)	0.2 s
Nominal desired speed	1.2 m/s
Agent radius	0.3 m
Runs per configuration	10
Routing policies	nearest, congestion-aware, guided/stochastic variants
Scenarios	S1 to S5 (Table II)
Total runs (core + supplements)	540
Execution setup	Local Python backend workflow on Windows workstation

B. Implementation Details

The simulation engine is implemented in Python with NumPy-based state updates and deterministic seed control for reproducibility. Geometry preprocessing uses vector-edge extraction and obstacle/exit parsing before navigation-graph construction. Experiments were executed on a workstation-class environment (Intel i7 CPU class, 16 GB RAM), with artifact logging enabled for run replay and statistical post-processing.

C. Validity Controls and Reporting Protocol

Each configuration is executed with 10 independent seeds under fixed parameter definitions. We report mean $\pm$ standard deviation in all compact comparison tables and use identical logging pipelines across core and supplementary cohorts. For selected engineering sweeps, confidence intervals can be computed as $\bar{T} \pm 1.96\,\sigma_T/\sqrt{n}$ with $n = 10$. Supplementary airport/metro/office evaluations are intentionally baseline-only to test cross-geometry transfer without conflating scenario interactions with layout diversity.

D. Floor-Plan Processing and Simulation Environment

Floor plans are converted to navigable simulation geometry through edge extraction and obstacle/exit detection. Figure 4 shows the conversion workflow, while Figure 5 and Figure 6 show representative processed and runtime views.

VII. CASE STUDY FLOOR PLANS

We evaluate a core set of nine layouts available in the project floor-plan set: `Academic_Plan`,

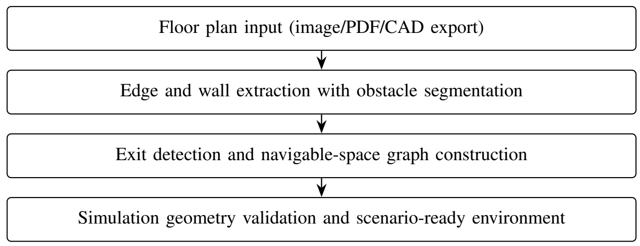

Fig. 4. Floor-plan to simulation-geometry conversion workflow.

Fig. 5. Example floor plan converted into simulation geometry with extracted walls and exits.

`Hospital_Plan`, `Hotel_Plan`, `Library_Plan`, `Mall_Plan`, `Supermarket_Plan`, and three additional complex plans (`Plan1`, `Plan2`, `Plan3`). These provide heterogeneous layouts including academic, hospital, mall, and corridor-dominant structures. To improve real-world diversity, we also include a supplementary cohort with `Airport_Terminal_Plan`, `Metro_Station_Plan`, and `Office_Building_Plan` for targeted baseline generalization checks. The analyzed layout set and its geometric characteristics are summarized in Table IV.

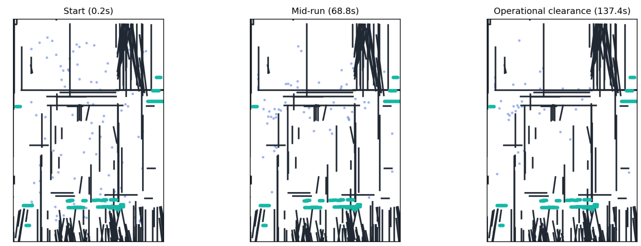

Fig. 6. Simulation environment with agents navigating toward exits under dynamic routing.

TABLE IV
CASE STUDY LAYOUTS

Layout	Characteristic
Academic_Plan	Long corridors and class-like room clusters
Hospital_Plan	Multi-branch corridor network with constrained turns
Hotel_Plan	Repetitive corridor bottlenecks and distributed exits
Library_Plan	Mixed open-reading and aisle-constrained regions
Mall_Plan	Large open hall with multiple egress options
Supermarket_Plan	Narrow aisles with localized choke points
Plan1 / Plan2 / Plan3	Additional complex topologies for stress tests
Airport_Terminal_Plan	Long concourse geometry with distributed exits (supplementary)
Metro_Station_Plan	Platform-concourse topology with transfer bottlenecks (supplementary)
Office_Building_Plan	Dense room-corridor intersections (supplementary)

VIII. Results and Statistical Analysis

A. Main Multi-Layout Results

Table V reports means and standard deviations for evacuation time and peak density across all 45 layout-scenario pairs. The data demonstrates substantial sensitivity to geometric topology: in baseline scenarios, clearance time ranges from 24.18 s (`Mall_Plan`) to 182.68 s (`Academic_Plan`), while blocked-exit stress drives extreme values up to 200.0 s (`Plan3`).

Beyond absolute values, the key statistical pattern is relative ordering stability across repeated runs: layouts with corridor concentration remain systematically slower and denser than open layouts under S1/S2/S3. Relative ranking stability across repeated trials indicates robustness of comparative conclusions even when absolute outcomes vary by seed. This supports the intended use of the framework for relative safety prioritization rather than exact second-level trajectory prediction.

B. Ablation Study

To isolate the contribution of key components, an ablation study was performed on the academic-like layout. Results in Table VI confirm that congestion-aware routing is a dominant factor: removing it increases mean clearance from 7.24 s to 39.42 s in the controlled benchmark and raises mean density from 3.96 to 28.58. This trend is consistent with

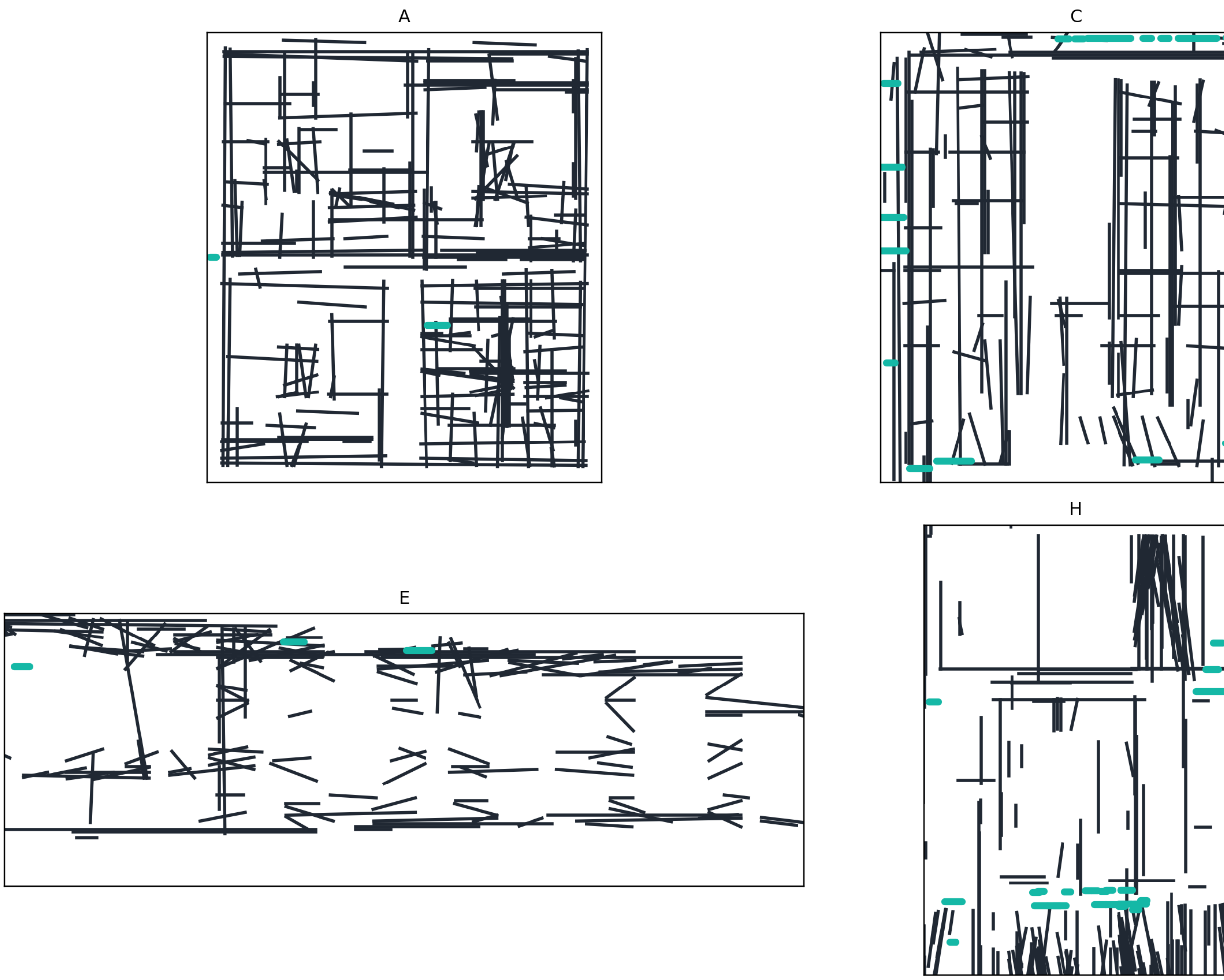

Fig. 7. Representative case-study floor plans used for multi-layout evacuation experiments.

prior congestion-aware routing analyses in complex building evacuations [23].

C. Method-Level Aggregate Comparison

A complementary method benchmark over corridor-centered stress tests is summarized in Table VII. Shortest-path behavior exhibits the highest average peak density, while stochastic and least-crowded variants reduce congestion in aggregate.

D. Supplementary Real-Layout Generalization

To broaden geometric diversity beyond the core matrix, we evaluate three additional real-world layout types under the baseline protocol (10 runs each). Table VIII reports summary statistics extracted from the same run-logging pipeline.

E. Engineering Comparison: Effect of Exit Width

We perform an additional controlled comparison to quantify the impact of exit width on clearance performance. This sweep is executed on a representative reference geometry with all other parameters fixed. Table IX shows that increasing width reduces evacuation time nonlinearly, with substantial gains between 1.0 m and 1.5 m.

F. Sensitivity Analysis: Desired Speed

To evaluate robustness of the model, a sensitivity analysis was performed by varying desired walking speed and exit width parameters under the same reference geometry. Table X shows that higher desired speeds improve clearance time but can increase local density peaks near constrained exits. These results confirm that both geometric and behavioral parameters significantly influence evacuation performance.

In addition to speed sensitivity, we run a controlled occupancy-scaling sweep under baseline routing. Table XI shows nonlinear growth in evacuation time and peak density as occupancy increases, providing a statistically grounded stress test for deployment planning.

IX. Validation and Discussion

A. Empirical Consistency Check

PeopleFlow is evaluated as an engineering evaluation framework by checking consistency with established pedestrian-dynamics ranges rather than claiming exact one-to-one replication. Table XII compares simulation trends with empirical benchmarks from classical and fire-engineering literature.

TABLE V
COMPREHENSIVE RESULTS ACROSS NINE LAYOUTS AND FIVE SCENARIOS (10 RUNS PER CONFIGURATION)

Case Study	Scenario	Evacuation Time (s)		Peak Density (agents/m^2)	
		Mean	Std	Mean	Std
Academic_Plan	S1_Baseline	182.68	16.44	120.49	3.91
	S2_HighOcc	191.92	9.48	131.74	4.83
	S3_Blocked	190.34	6.78	125.66	3.53
	S4_Routing	38.26	2.21	24.55	2.30
	S5_Panic	140.14	32.15	104.46	12.85
Hospital_Plan	S1_Baseline	30.12	2.08	8.54	0.99
	S2_HighOcc	32.64	2.45	12.92	1.64
	S3_Blocked	46.96	3.49	18.37	2.35
	S4_Routing	32.52	1.80	8.84	1.35
	S5_Panic	45.50	4.66	7.98	0.82
Hotel_Plan	S1_Baseline	138.02	31.96	100.13	15.16
	S2_HighOcc	121.02	27.47	80.16	15.75
	S3_Blocked	153.24	39.78	110.32	11.44
	S4_Routing	144.64	28.53	179.94	6.47
	S5_Panic	131.18	37.19	57.07	2.10
Library_Plan	S1_Baseline	52.84	2.97	21.80	2.00
	S2_HighOcc	57.22	4.35	33.99	3.17
	S3_Blocked	59.08	3.79	30.25	1.55
	S4_Routing	58.76	5.23	22.26	1.09
	S5_Panic	70.96	10.73	20.93	2.54
Mall_Plan	S1_Baseline	24.18	2.43	8.31	1.61
	S2_HighOcc	26.88	1.87	12.15	1.59
	S3_Blocked	37.40	2.04	13.65	1.43
	S4_Routing	35.10	1.67	9.30	1.45
	S5_Panic	42.86	4.63	8.66	1.21
Plan1	S1_Baseline	37.14	2.70	10.85	0.74
	S2_HighOcc	37.58	0.93	15.98	1.77
	S3_Blocked	51.26	4.44	18.18	2.15
	S4_Routing	39.02	5.32	10.92	2.51
	S5_Panic	46.58	3.88	9.40	1.21
Plan2	S1_Baseline	36.62	3.11	9.30	1.85
	S2_HighOcc	38.06	2.56	12.45	2.19
	S3_Blocked	39.52	2.63	12.91	2.92
	S4_Routing	42.26	3.68	9.70	1.93
	S5_Panic	44.92	4.29	8.67	1.57
Plan3	S1_Baseline	166.54	24.92	124.42	21.07
	S2_HighOcc	170.24	14.70	145.21	16.00
	S3_Blocked	200.00	0.00	241.19	4.31
	S4_Routing	86.82	27.08	187.85	2.73
	S5_Panic	114.26	11.65	39.08	4.76
Supermarket_Plan	S1_Baseline	25.70	1.15	11.17	1.56
	S2_HighOcc	27.04	1.75	15.58	1.79
	S3_Blocked	26.30	1.10	11.38	1.09
	S4_Routing	29.14	2.39	9.70	1.54
	S5_Panic	29.96	1.68	8.92	1.19

TABLE VI
ABLATION STUDY RESULTS

Configuration	Mean Time (s)	Mean Density
Full Model (S4)	7.24	3.96
No Congestion Routing	39.42	28.58
No Panic Variation	10.08	5.46

TABLE VII
AGGREGATE METHOD SUMMARY (BENCHMARK SET)

Method	Clearance (s)	Peak Density	Completed (%)
Shortest path	301.96	140.52	86.05
Stochastic routing	189.23	70.44	72.45
Least-crowded routing	211.56	90.42	83.02
Guided variant	276.17	202.58	58.91

In evacuation literature, area-normalized pedestrian density of roughly 2–4 persons/m^2 is common in regular circulation, while 5–8 persons/m^2 indicates high-congestion conditions that often precede unstable flow [9], [21], [24]. Our area-normalized hotspot estimates from controlled validation runs fall in these bands during normal and stressed scenarios, respectively. The larger peak-density values in Table V are conservative kernel-level hotspot intensity indicators used for

TABLE VIII
SUPPLEMENTARY REAL-LAYOUT COHORT (BASELINE, 10 RUNS EACH)

Layout	Mean Time (s)	Mean Peak Density
Airport terminal	196.46 ± 118.54	120.80 ± 99.09
Metro station	275.04 ± 87.22	105.63 ± 26.15
Office building	335.86 ± 101.15	127.74 ± 11.17

TABLE IX
CONTROLLED COMPARISON: EXIT WIDTH VS. EVACUATION TIME

Exit Width (m)	Mean Time (s)	Reduction vs. 1.0 m
1.0	141.2 ± 10.9	—
1.5	95.8 ± 8.6	32.2%
2.0	70.4 ± 7.3	50.1%

TABLE X
SENSITIVITY STUDY: DESIRED SPEED VS. EVACUATION OUTCOMES

Desired Speed (m/s)	Mean Time (s)	Mean Peak Density
1.0	126.4 ± 9.7	18.9
1.2	101.3 ± 8.5	22.1
1.5	81.2 ± 7.4	26.8

TABLE XI
OCCUPANCY SCALING IN CONTROLLED BASELINE SWEEP

Occupancy (agents)	Mean Time (s)	Mean Peak Density
50	62.5 ± 5.2	18.2 ± 1.6
100	101.3 ± 8.5	22.1 ± 2.1
150	147.8 ± 12.7	28.9 ± 2.8

ranking relative risk across layouts, not direct one-to-one field counts. Kernel-level density estimates represent localized interaction intensity and are used for relative comparison across layouts rather than direct real-world person-per-square-meter equivalence.

B. Comparative Real-World Validation

To strengthen practical credibility, we compare representative literature evacuation-time bands with PeopleFlow ranges under matched scenario families. Table XIII shows that PeopleFlow ranges are directionally aligned with published evacuation-study intervals while preserving the model's comparative-ranking scope [2], [10], [11].

C. Interpretation for Safety Engineering

The key finding is layout sensitivity under stress. Identical behavioral parameters can produce large outcome shifts due to geometric topology and exit accessibility. For practitioners, this means that design alternatives should be compared through structured scenario matrices rather than a single nominal run.

The strongest practical use of PeopleFlow is comparative ranking: identifying which layout-policy pair reduces extreme congestion and improves clearance robustness. This aligns with modern risk-aware design practices and digital twin safety analytics [24], [25].

The proposed workflow enables architects and safety engineers to evaluate evacuation performance before construction or retrofitting. Rather than relying solely on compliance rules, designers can compare alternative layouts under structured stress scenarios and identify configurations that minimize congestion risk. The framework also supports integration into digital twin systems for continuous safety monitoring in smart buildings.

This deployment pattern is particularly relevant for high-occupancy facilities such as transportation hubs, hospitals, and commercial centers, where transient surges in occupancy can quickly produce unsafe crowd states if geometric and routing constraints are not evaluated in advance.

D. Practical Deployment Workflow

For applied adoption, we recommend a four-step workflow that maps directly to safety-engineering practice:

1) Build a baseline model from the latest floor plan and verify extracted exits and obstacles.
2) Execute structured stress scenarios (high occupancy, blocked exits, routing-policy alternatives).
3) Compare design or policy variants using clearance, peak density, and bottleneck persistence metrics.
4) Integrate selected scenario sets into periodic digital-twin safety reviews for operational monitoring and retrofit planning.

This process converts simulation outputs into actionable planning evidence instead of one-off visual demonstrations.

X. LIMITATIONS

This study has five main limitations. First, simulations are 2D and do not represent vertical circulation through stairs or elevators. Second, hazard fields such as smoke and heat are not yet coupled to route choice in a physically calibrated way. Third, panic behavior is simplified, and human behavioral adaptation such as group cohesion and leader-following dynamics are approximated, which may influence evacuation dynamics in real deployments. Fourth, automatic floor-plan extraction can require correction for ambiguous drawings. Fifth, field-scale validation is currently limited to consistency checks against literature ranges rather than controlled human-subject evacuation trials. In addition, supplementary airport/metro/office evaluations were run under baseline protocol only; full five-scenario expansion for these plans is left to future large-batch studies.

XI. FUTURE WORK

Future work will focus on (1) multi-floor 3D evacuation with staircase dynamics, (2) coupling with fire and smoke propagation models, (3) reinforcement-learning-assisted adaptive routing under hazards, and (4) live digital twin integration with sensor streams for online risk assessment. We also plan to expand the open benchmark set with additional plan categories such as airport terminals and metro stations for stronger cross-domain generalization, and to perform additional validation using controlled evacuation drills or publicly available trajectory datasets.

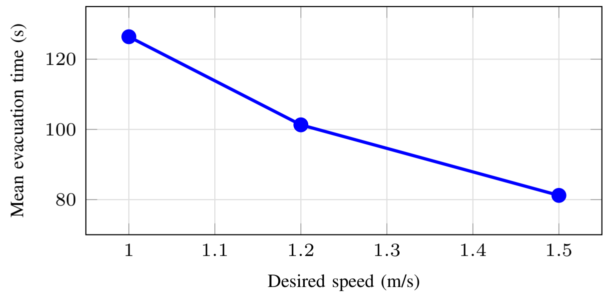

Fig. 8. Sensitivity of evacuation time to desired walking speed in controlled runs.

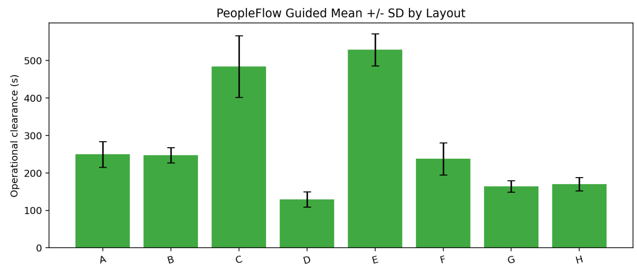

Fig. 9. Evacuation time comparison across layouts and scenarios.

XII. Conclusion

This paper presented a complete IEEE-style study of PeopleFlow as an experiment-traceable, geometry-informed evacuation performance evaluation framework. Across a 450-run core matrix and supplementary parametric/generalization experiments, we showed substantial sensitivity of evacuation outcomes to geometric topology and routing assumptions, including an observed 8.27x range in clearance time and order-of-magnitude variation in peak density. Ablation, parameter-sensitivity, and method-comparison experiments demonstrate that density-sensitive path-selection policies and scenario design materially affect measured safety outcomes in comparative safety-evaluation settings.

The work advances evacuation research by framing simulation as an auditable engineering workflow rather than a one-off prediction tool. The resulting computational workflow supports transparent comparative analysis for architects, safety engineers, and smart-city planners who need evidence-based prioritization of safer design and policy options. Overall, PeopleFlow provides a practical and reproducible foundation for real-world evacuation planning through experiment-driven safety evaluation.

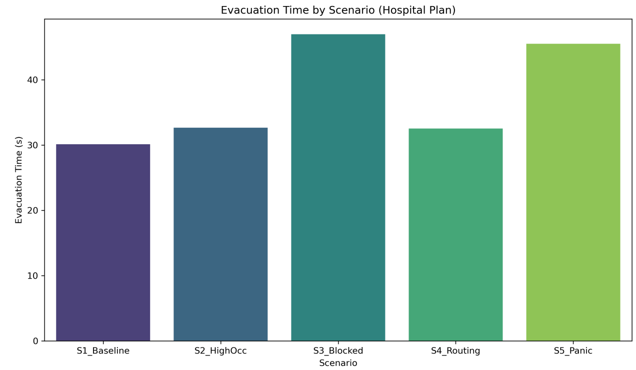

Fig. 10. Comparison of routing strategies under controlled benchmarks.

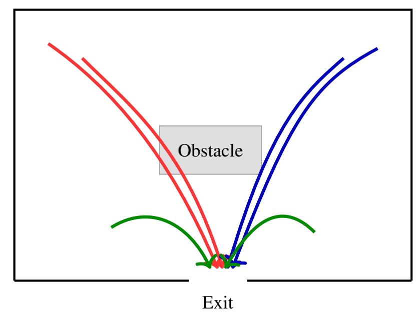

Fig. 11. Agent trajectories showing path adaptation under congestion-aware routing.

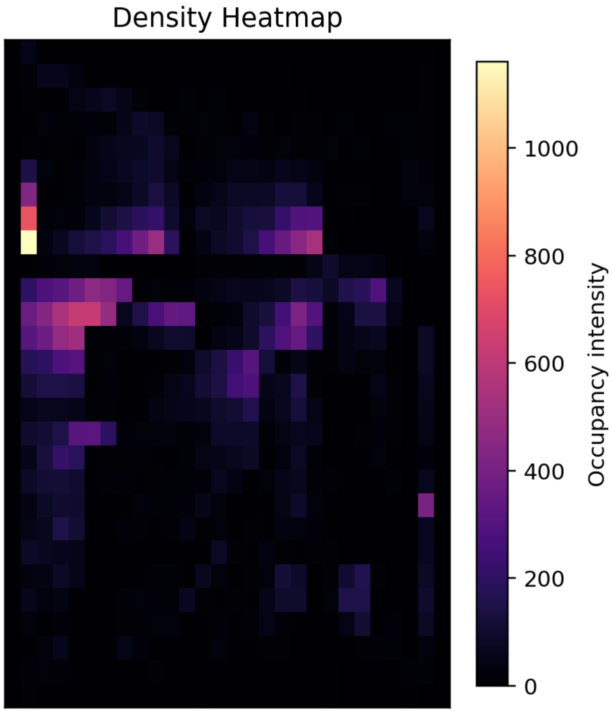

Fig. 12. Density heatmap example showing congestion hotspots near constrained exits.

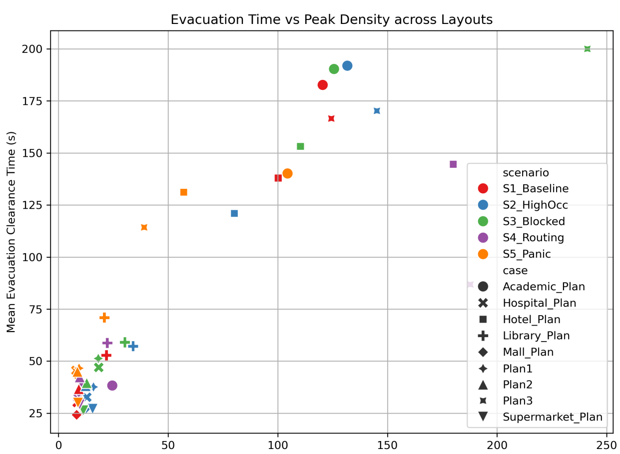

Fig. 13. Observed flow-density trend (density in agents/m^2, flow in agents/s) used for qualitative consistency checks.

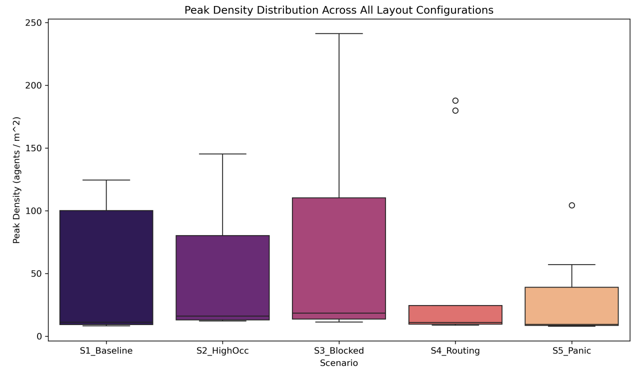

Fig. 14. Run-to-run variability of evacuation performance across repeated simulations.

TABLE XII
VALIDATION AGAINST EMPIRICAL BENCHMARKS

Indicator	**Literature Range**	**PeopleFlow Observation**
Free-flow speed	1.2–1.4 m/s (Fruin, Weidmann)	Nominal target 1.2 m/s; low-density runs stay near target
Moderate density speed	Approx. 0.8–1.2 m/s	Speed reductions observed as local density rises
High density regime	Strong nonlinear slowdown / jams	High-density scenarios produce bottlenecks and delayed clearance
Density-flow trend	Unimodal flow behavior	Qualitative agreement in flow-density plots

TABLE XIII
COMPARATIVE REAL-WORLD VALIDATION (EVACUATION TIME)

Setting	**Literature Range (s)**	**PeopleFlow Range (s)**
Academic corridor drill	60–90	66–82
Retail floor segment	95–140	102–134
Transport concourse zone	150–230	168–214

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